

EXAM 7 – SPRING 2014

10. (1 point)

a. (0.5 point)

Within the context of a stochastic framework for loss reserving, briefly describe two types of expert opinion that can be used to override the calculated parameters in a predictive reserving model.

b. (0.5 point)

Given the following parameters and means of three Gamma distributions:

α_i	β_i	<u>Mean</u>
30,000	12.00	2,500
3,000	1.20	2,500
300	0.12	2,500

Discuss how the β_i value influences the trade-off between the chain-ladder and Bornhuetter-Ferguson estimates in a Bayesian model.

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SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 10	
TOTAL POINT VALUE: 1	LEARNING OBJECTIVE: A8: Identify data issues and related model adjustments for reserving models.
SAMPLE ANSWERS	
Part a: 0.5 point	
<i>Sample Answer 1</i> - Can incorporate an a priori opinion regarding the expected ultimate level of losses => this could come from an underwriter (expert), based on his or her knowledge about a change in the quality of risks written (expect experience to be better or worse than in the past). - Could incorporate a prior opinion regarding the loss development factors => here claims manager may be the expert, as he or she may have knowledge about a change to the claims process that would make selected LDFs more appropriate than those calculated in the model. <i>Sample Answer 2</i> - The LDF for a certain year and development period can be over-ridden based on expert opinion regarding recent claim process changes. - The LDF along a diagonal can be overridden based upon expert opinion regarding a calendar year effect such as legislative changes impacting all claims after a certain calendar date. <i>Sample Answer 3</i> - Can apply expert opinion on a column parameter such as an LDF. Can judgementally pick higher or lower factor based on known changes, such as change in claims philosophy. - Can apply expert opinion on row parameters such as the predicted ultimate loss. Can pick higher or lower estimate based on known changes within the company. <i>Other accepted answers:</i> - Actuarial can override LDF factors if not reflective of current exposure set or add additional trend factors if change in inflation, trend, claim litigation will arise in the future.	
Part b: 0.5 point	
<i>Sample Answer 1</i> For a gamma distribution, the variance is α/β^2. Choose a small β, get high variance, less weight to B-F estimate and more to chain ladder. Choose a large β, get low variance, more weight to B-F estimate and less to chain ladder.	

SAMPLE ANSWERS AND EXAMINER'S REPORT

Sample Answer 2

The credibility factor Z that gets applied to the chain ladder method is $Z = P/(\beta\phi + P)$. With B being in the denominator, large values of β will lead to small values of Z , and thus less reliance on the chain ladder method. So the first distribution will rely more on the BF than the second, and the second more than the third.

EXAMINER'S REPORT

Part a

The candidate was expected to identify two types of expert opinion that could be used to modify model results.

In order to achieve full credit, the candidate had to identify an issue that an expert may have commentary on (e.g., a claims manager might have input on claim process changes) and how that opinion would/could impact the model (e.g., judgmentally select an LDF rather than relying on model output.)

Many candidates received only partial credit because they did not clearly identify the expert/expert's view but limited their answer to what model parameter would change (e.g., "Could make manual adjustments to empirical age-to-age factors.")

Part b

The candidate was expected to understand the relationship between the β parameter and the relative weight given to the BF vs. Chain-ladder methodologies.

The majority of candidates received full credit for this part.

Generally, candidates who did not receive full credit either had the relationship reversed or simply failed to attempt the question. There were a few candidates who discussed the β parameter and its relationship to credibility and/or variance but did not state the final effect on the weighting of the BF and Chain Ladder.